Study notes: Section 2: charge in C-R and L-C circuits; leakage (Murugeshan 12.3 to 12.5)

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

12.3 Growth and decay of charge in a circuit with resistance and capacitance

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 315–317 (printed), with Example 1 (capacitor charged through 2 MΩ) and Examples 2 and 3. Syllabus items 15 and 16. Equations (a)(1)–(a)(3) and (b)(1)–(b)(3) are given in full.

Before you start

- Capacitance and charge, $Q = CV$ (Plus Two; pre-physics P3).
- Ohm's law (pre-physics P1) and the first-order equation solved in §12.1 of these notes.

1. Growth of charge (charging)

A capacitor C and a resistance R are joined to a cell of emf E through a Morse key K (Fig. 12.6). When the key is pressed, a momentary current I flows through R. Let Q be the charge on the capacitor at time t. The p.d. across the capacitor is Q/C and the p.d. across the resistor is RI.

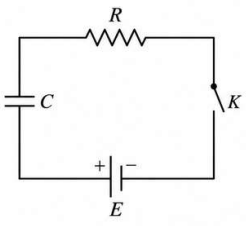


Fig. 12.6 (book): capacitor C, resistance R, Morse key K and cell E in series.

The emf equation of the circuit is

$$E = \frac{Q}{C} + RI \qquad \text{(12.3(a)(1))}$$

Since $I = dQ/dt$, this is $E = Q/C + R\,dQ/dt$.

2. The final charge

The charge grows until it reaches its maximum Q_0. At that instant $I = dQ/dt = 0$ and the p.d. across the capacitor equals the emf: $E = Q_0/C$.

3. Separating the variables

Substitute $E = Q_0/C$ into the equation, so $Q_0/C = Q/C + R\,dQ/dt$. Multiply by C:

$$(Q_0 - Q) = CR\frac{dQ}{dt}$$
$$\frac{dQ}{Q_0 - Q} = \frac{dt}{CR} \qquad \text{(12.3(a)(2))}$$

4. Integration and boundary condition

Integrating gives $-\log_e(Q_0 - Q) = t/CR + K$. At $t = 0$, $Q = 0$, so $K = -\log_e Q_0$. Then $\log_e \frac{Q_0 - Q}{Q_0} = -\frac{t}{CR}$, which gives

$$Q = Q_0\left(1 - e^{-t/CR}\right) \qquad \text{(12.3(a)(3))}$$

5. Time constant

The time constant is $\tau = CR$. At $t = CR$, $Q = Q_0(1 - e^{-1}) = 0.632\,Q_0$. The time constant is therefore the time for the charge to reach 0.632 of its maximum. The rate of growth is $dQ/dt = (Q_0/CR)e^{-t/CR}$. A smaller product CR gives faster charging. The rate of charging is greatest at the start and falls as Q approaches Q_0 (Fig. 12.7).

6. Decay of charge (discharging)

Now release the Morse key so that the capacitor, charged to Q_0, discharges through R. The charge flows out and forms a current. Here $E = 0$, so

$$R\frac{dQ}{dt} + \frac{Q}{C} = 0 \qquad \text{(12.3(b)(1))}$$

Separating the variables and integrating with $Q = Q_0$ at $t = 0$ gives

$$Q = Q_0 e^{-t/CR} \qquad (12.3(b)(2))$$

The charge decays exponentially and becomes zero only after an infinite time. The rate of discharge is

$$I = \frac{dQ}{dt} = -\frac{Q_0}{CR} e^{-t/CR} = -\frac{Q}{CR} \qquad (12.3(b)(3))$$

In Eq. (b)(2), at $t = CR$, $Q = Q_0 e^{-1} = 0.368\, Q_0$. So the time constant is also the time for the charge to fall to $1/e$ of its maximum. Growth and decay are shown together in Fig. 12.7.

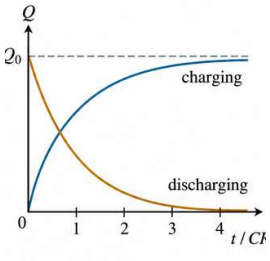


Fig. 12.7 (book): charge against time; growth to Q_0 and discharge from Q_0 are complementary.

7. Worked example 1 (printed p. 316)

A capacitor is charged through a resistance of $2\ \text{M}\Omega$. It takes 0.5 s to reach three quarters of its final charge. From Eq. (a)(3), $\frac{3}{4} = 1 - e^{-0.5/CR}$, so $e^{0.5/CR} = 4$ and $C = 0.5/(2 \times 10^6 \ln 4) = 0.18\ \mu\text{F}$.

8. Worked example 2 (printed p. 317)

A capacitor of $0.1\ \mu\text{F}$ is discharged through $10\ \text{M}\Omega$. The time for the p.d. to fall to half its value is $t = CR \ln 2 = 0.6931$ s, since $Q/Q_0 = V/V_0$ for the same capacitor.

12.4 Measurement of high resistance by leakage (background)

Background. *This section is outside the Module 4 items (15–22). It is kept because it uses the discharge law of §12.3 and the book gives it in full. No equations are numbered in the book for this section. Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 318–319 (printed).*

1. Principle

A capacitor of capacitance C and initial charge Q_0 is allowed to discharge through a high resistance R for a time t. From §12.3, the charge that remains is $Q = Q_0 e^{-t/CR}$. Therefore

$$\frac{Q_0}{Q} = e^{t/CR}$$

$$\Rightarrow$$

$$R = \frac{t}{C\log_e(Q_0/Q)} = \frac{t}{2.3026\,C\log_{10}(Q_0/Q)}$$

If R is high, CR is large and the discharge is very slow. Measuring Q_0/Q then gives R.

2. Circuit and procedure

C is a capacitor of known value, R is the high resistance, B.G. is a ballistic galvanometer, E is a cell and K_1, K_2, K_3 are tap keys (Fig. 12.8).

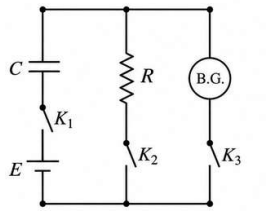


Fig. 12.8 (book): capacitor C, high resistance R, ballistic galvanometer B.G., cell E and keys K1, K2, K3.

(*i*) With K_2 and K_3 open, charge C fully through K_1. Open K_1 and close K_3. The capacitor discharges through the galvanometer, which gives a throw θ_0, proportional to Q_0.

(*ii*) Charge C again, open K_1, and close K_2 for a known time t. Some charge leaks through R. Open K_2 and close K_3. The remaining charge Q gives a throw θ, proportional to Q.

Since $Q_0/Q = \theta_0/\theta$,

$$R = \frac{t}{2.3026\,C\log_{10}(\theta_0/\theta)}$$

3. Graph

A plot of $\log_{10}(\theta_0/\theta)$ against t is a straight line through the origin (Fig. 12.9). Its slope gives $\frac{t}{\log_{10}(\theta_0/\theta)}$, and with C known, R follows.

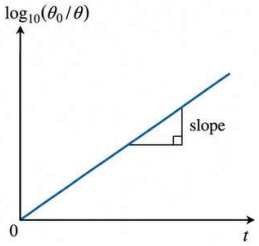


Fig. 12.9 (book): straight line of log10(θ0/θ) against t; the slope gives t / log10(θ0/θ).

4. Worked examples (printed pp. 318–319)

(1) For $C = 2\ \mu$F, $R = 100\ \text{M}\Omega$ and $Q_0/Q = 2$, the leakage time is $t = 2.3026\,CR\log_{10}2 = 138.7$ s. (2) For $C = 0.2\ \mu$F, a kick of 12 cm and then 6 cm after 2 s gives $R = 14.43\ \text{M}\Omega$.

12.5 Charging and discharging of a capacitor through an inductance

Source: book-sources/m4-s2-cr-lc.html, Murugeshan pp. 319–321 (printed). Syllabus items 15 and 16. The book numbers the charging equations (1)–(4) and, after a new heading, the discharge equations (1)–(4) again; these are tagged here as (i) and (ii). Section 12.6 onwards is outside the module and is not included.

Before you start

- The charge and the p.d. of a capacitor (pre-physics P3); self-inductance and back emf (pre-physics P2).
- Simple harmonic motion: $d^2x/dt^2 + \omega^2 x = 0$ (pre-math H2).

1. Charging a capacitor through an inductance

A capacitance C and an inductance L are joined to a cell E through a Morse key (Fig. 12.10). Pressing the key starts a momentary current through L and the capacitor begins to charge. Let Q be the charge on C at time t. The emf equation is

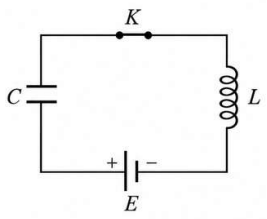


Fig. 12.10 (book): capacitor C and inductance L joined through a Morse key to a cell E.

$$L\frac{dI}{dt} + \frac{Q}{C} = E \qquad (12.5(i)(1))$$

Using $I = dQ/dt$, so that $dI/dt = d^2Q/dt^2$, and writing $Q_0 = CE$ for the maximum charge, the equation becomes

$$L\frac{d^2Q}{dt^2} + \frac{Q}{C} = E$$
$$\frac{d^2Q}{dt^2} + \frac{1}{LC}(Q - Q_0) = 0 \qquad (12.5(i)(2))$$

This is the equation of undamped simple harmonic motion, with period

$$T = \frac{2\pi}{\sqrt{1/LC}} = 2\pi\sqrt{LC} \qquad (12.5(i)(3))$$

The charging is oscillatory. The frequency is

$$\nu = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(i)(4))$$

2. Discharging a capacitor through an inductance

The capacitor is charged to Q_0 with the inductance out of the circuit. The key is then released so that C discharges through L (Fig. 12.11). No external emf acts during the discharge.

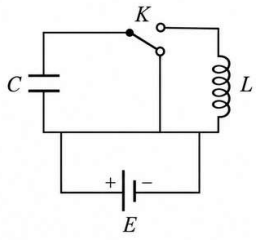


Fig. 12.11 (book): capacitor C discharging through inductance L with no external emf.

$$\frac{Q}{C} + L\frac{dI}{dt} = 0$$
$$\frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0 \qquad (12.5(ii)(1))$$

This is undamped simple harmonic motion. Put $1/LC = \omega^2$. The general solution is

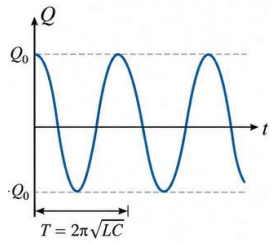
$$Q = A\sin(\omega t + \theta) \qquad (12.5(ii)(2))$$

with $\omega = 1/\sqrt{LC}$. At $t = 0$, $Q = Q_0$, which gives $Q_0 = A\sin\theta$. Taking $\theta = \pi/2$ gives $A = Q_0$, and

$$Q = Q_0 \sin\left(\frac{t}{\sqrt{LC}} + \frac{\pi}{2}\right) \qquad (12.5(ii)(3))$$

So Q decreases sinusoidally: the discharge is oscillatory. The natural frequency is

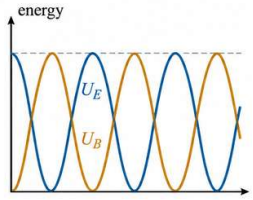
$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \qquad (12.5(ii)(4))$$



Undamped oscillation of charge, Q = Q₀ sin(t/√LC + π/2), period T = 2π√LC (book, §12.5).

3. Energy in the oscillation

When C is fully charged, all the energy is stored in the electric field of the capacitor. During discharge, a current flows through the coil for a short time and the energy is stored in the magnetic field of the coil. The energy then passes back and forth between the two fields, which is the basis of the oscillations used in wireless telegraphy.



Energy alternates between the electric field of C and the magnetic field of L (book, §12.5).